

Problem

The merged array should also be sorted in non-decreasing order. Complete the function mergeSortedArrays that takes the following parameters:

int arr1[n]: the first sorted array
int arr2[m]: the second sorted array
Returns int[]: the merged sorted array

Input Format

The first line contains an integer n, the size of the first array.

The second line contains n space-separated integers representing the first sorted array. The third line contains an integer m, the size of the second array. The fourth line contains m space-separated integers representing the second sorted array.

Constraints

$1 \leq n, m \leq 10^5$ $-10^9 \leq arr1[i], arr2[i] \leq 10^9$ Both arrays are sorted in non-decreasing order.

Output Format

Print the merged sorted array as space-separated integers.

Sample Input 0

```
4
1 3 5 7
4
2 4 6 8
```

Change Theme

Language

Java 15



```
1 import java.io.*;
2 import java.util.Scanner;
3 class Solution{
4     public static void main(String[] args){
5         Scanner sc = new Scanner(System.in);
6         int n = sc.nextInt();
7         int[] arr = new int[n];
8         int i=0;
9         while(i<n){
10             arr[i] = sc.nextInt();
11             i++;
12         }
13         int m = sc.nextInt();
14         int[] arr2 = new int[m];
15         int j=0;
16         while(j<m){
17             arr2[j] = sc.nextInt();
18             j++;
19         }
20         int[] merged = new int[n+m];
21         i=0;
22         j=0;
23         int k=0;
24         while(i<n && j<m){
25             if(arr[i]<=arr2[j]){
26                 merged[k] = arr[i];
27                 i++;
28             }else{
29                 merged[k] = arr2[j];
30                 j++;
31             }
```